

C Essentials

Independent study sheet • Not official DPS Agra material • No personal information required

1. First program

```
#include <stdio.h>

int main(void) {
    printf("Hello C\n");
    return 0;
}
```

Compile using a sandboxed compiler; C code must not run directly on your school API server.

2. Values and branches

```
int score = 70;
if (score >= 50) {
    puts("Pass");
}
```

Integer arithmetic truncates; choose types deliberately.

3. Safe input

```
char name[32];
if (fgets(name, sizeof name, stdin)) {
    printf("%s", name);
}
```

Avoid unbounded input functions and uninitialized pointers.

TRY IT YOURSELF

1. Change program to print two lines.
2. Write a simple even/odd condition.
3. Explain why bounded input matters.

Open the portal Practice IDE for Python, SQL, Web and Block coding.

For other languages, use an approved isolated compiler. Verify unfamiliar code before running.